Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/08/2007
PI Name:	Maurer, Richard H. Ph.D.		
Project Title:	Combined Ion and Neutron Spectrometer for Space Applications		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Technology Development Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	20723-6099	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	08/01/2004	**End Date:**	07/31/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Goldsten, John (The Johns Hopkins University Applied Physics Laboratory) Roth, David (The Johns Hopkins University Applied Physics Laboratory) Zeitlin, Cary (Lawrence Berkeley National Laboratory) Haggerty, Dennis (The Johns Hopkins University Applied Physics Laboratory)		
Grant/Contract No.:	NCC 9-58-TD00404		
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Task Description:	We proposed to design and fabricate a Combined Ion and Neutron Spectrometer (CINS) for Space Applications. This instrument will improve upon existing charged particle species and energy spectrometers (e.g., MARIE on Mars Odyssey) and incorporate the capability of the previously developed NSBRI Neutron Energy Spectrometer to yield a single and complete ionizing radiation environment monitor for application in space habitats and transport vehicles. An improved charged particle/ion telescope detector system is being designed at Lawrence Berkeley National Laboratory and JHU/APL. It will be combined with the NSBRI Neutron Spectrometer at JHU/APL into a single instrument for ground based research at accelerators. CINS will be used to compare its energy spectra with the Tissue Equivalent Proportional Counter (TEPC) to investigate basic hypotheses about 1) the lack of radiation environment information from the TEPC LET (Linear Energy Transfer) spectrometer due to detection limitations with respect to high energy protons, low energy heavy ions and neutrons; and 2) the increased risk due to the multiplication of primary beams of protons and heavy ions by spacecraft materials producing a secondary particle environment of protons, neutrons and heavy ion fragments. The originally proposed objectives were refined with respect to specific metrics in February/March 2005 to 1) eliminate the gain saturation of MARIE for heavy ions with LET > 35 keV/micron; 2) increase the dynamic range vs. the MARIE instrument by a factor of 10-20 to a minimum of 1000:1 to include protons with energies above 100 MeV; 3) increase the maximum instrument event detection rate by at least a factor of 10 to a minimum of 30 Hz. We have enhanced these original objectives by choosing the more ambitious goals of 4) a 10,000:1 dynamic range; 5) a 100 Hz event rate. The end product of the effort will be a Countermeasure Readiness Level 6 instrument. The conceptual design of the CINS instrument is facilitated through the use of Monte Carlo simulations that model the basic detector stack, establish the energy resolution, and quantify the issues that will affect the instrument's efficiency. For this effort we are using the GEANT4 toolkit. Our CINS design consists of a stack of 7 cylindrical detectors with a radius of 20 mm. There are 3 different types of detectors in the CINS stack. One is a standard Silicon (Si)solid state detector. The remaining two detector types are scintillators, one is similar to BC430 while the other scintillator is BGO. The first detector in the stack is a 1 mm thick BC430 detector followed by a 5 mm thick Si detector, then a second 1 mm BC430 detector, followed by three 5 mm Si detectors. The final detector is a 30 mm thick detector of BGO. The entire stack is wrapped in a 3mm thick Titanium shell. For the CINS system to accurately resolve solar energetic particle (SEP) events the system must cleanly separate the protons from the electrons. We simulated spectra of electrons to 20 MeV and protons to 1 GeV. While a small percentage of the electrons and protons do overlap in this simulation, the vast majority of the protons can be separated from the electrons. Protons up to ~150 MeV are stopped in the stack while protons with E>150 MeV penetrate through the system. Given reasonable energy resolution of the BGO this simulation shows that protons up to ~300 MeV can be uniquely identified. Helium spectra up to 750 MeV/nucleon and O, Si, and Fe spectra well past their GCR maxima (1 GeV/nucleon) are cleanly resolved. We re-drifted eight Si(Li) detectors for charged particles and neutrons from LBNL's existing stock. The second step was the creation of a guard ring accomplished by routing out the detector in a circular pattern at a fixed radius. These eight detectors have been evaluated in experiments at HIMAc and NSRL. Experiments were executed at NSRL in March 2006 by colliding a 1 GeV/amu iron beam with a 5 inch thick aluminum target simulating a large space vehcle. The progeny of these collisions, both charged partilce fragments and neutrons were monitored by the refurbished 5mm thick silicon detectors. For the Neutron Energy Spectrometer JHU/APL has designed and procured a plastic scintillator detector from Eljen for medium energy 1-14 MeV neutrons that have the maximum dose equivalent weighting factors of 10-20. This scintillator is now being mated to a photomultiplier tube. Schedule The overall schedule from the proposal can be divided into three phases of duration of 16 months each-- August 04-November 05--instrument design including modeling and fabrication December 05-March 07--finish fabrication, calibration and evaluation of detector systems; detector technology experiments at NSRL April 07-July 08--accelerator experiments with spacecraft shielding configurations and comparison with TEPC LET spectrometer for the charged particle and neutron environment. Schedule Details 2006: A) Procure and assemble the BGO scintillator for the charged particle telescope; B) Assemble the Eljen medium energy neutron detector system; C) Perform medium energy neutron detector evaluations at Columbia RARAF; D) Publish results of the March 2006 NSRL thick silicon detector experiments. 2007: A) BGO scintillator evaluation experiments at NSRL (Spring 07); B) Medium energy neutron collision experiments at NSRL (Spring 07); C) First experiments with the assembled charged particle telescope at NSRL (Summer 07).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Neutron detection for homeland security is of interest in the areas of 1) the location of radiological weapons used by terrorists, 2) scanning and inspecting imported cargo in ship holds or on the dock side and 3) smart portals for the scanning of personnel or small packages at selected points of entry.

Task Progress:	Charged particle telescope modeling and detector selection The conceptual design of the CINS instrument was facilitated through the use of Monte Carlo simulations that model the basic detector stack, establish the practical dynamic range and energy resolution, and quantify some of the issues that will affect the instrument's efficiency. For this effort the GEANT4 toolkit was employed. The CINS design consists of a stack of 7 cylindrical detectors with a radius of 20 mm. There are 3 different types of detectors in the CINS stack. There are four standard Silicon (Si) solid state detectors; two thin scintillators similar to BC430 and one thick BGO scintillator. The entire stack will be wrapped in a 3mm thick Titanium shell. Initial simulations for the CINS system show that while a small percentage of the electrons and protons do overlap,the vast majority of the protons can be separated from the electrons. Protons up to ~150 MeV are stopped in the stack while protons with E>150 MeV penetrate through the system yielding the characteristic folding of spectra in the dE/E presentation. Given reasonable energy resolution of the BGO this simulation shows that protons up to ~300 MeV can be uniquely identified while protons with E>300 MeV will likely be grouped as a single integral channel. Helium spectra up to 750 MeV/nucleon (the approximate GCR maxima for He), and O, Si, and Fe spectra well past their GCR maxima (~ 1 GeV/nucleon) are cleanly resolved. Eight 4cm X 5mm Si(Li) detectors have been refurbished by having the material re-drifted and having guard rings installed at LBNL. These detectors were evaluated for both charged particles and high energy neutron detection properties at CHIBA in February 2006 and at NSRL in March 2006. The experimental work at NSRL included the collision of the 1 GeV/amu iron beam with a 5 inch thick stack of aluminum. Four detector set-ups, two for charged particle fragments and two for neutrons simultaneously monitored the the progeny fo the collisions at several angles. The data on the charged particle and neutron energy spectra are presently being analyzed. A boron loaded plastic scintillator detector for medium energy neutrons has been designed and procured by APL. It is currently being assembled for use by being mated to a photomultiplier tube. The BGO scintillator for the charged particle stack has been designed and is being purchased. A final major task was the completion of our Balloon Borne Neutron Spectrometer for an Antarctic balloon flight for Marshall Space Flight Center using detector hardware from previous NSBRI grants and modeling results form the present grant.
Bibliography Type:	Description: (Last Updated: 04/27/2011)
Articles in Peer-reviewed Journals	Maurer RH, Kinnison JD, Roth DR. "Neutron production from 200-500 MeV proton interaction with spacecraft materials." Radiat Prot Dosimetry. 2005;116(1-4 Pt 2):125-30. PMID: 16604612 , Jan-2005
Articles in Peer-reviewed Journals	Maurer RH, Roth DR, Kinnison JD, Haggerty DK, Goldsten JO. "The NSBRI/APL Neutron Energy Spectrometer." Johns Hopkins APL Technical Digest. In Press, January 2006.. , Jan-2006
Papers from Meeting Proceedings	Maurer RH, Zeitlin CJ, Haggerty DK, Roth DR, Goldsten JO. "Compact ion and neutron spectrometer (CINS) for space applications." 2005 IEEE Nuclear Science Symposium Conference, Puerto Rico, October 23-29, 2005. In: 2005 IEEE Nuclear Science Symposium Conference Record. 2005 November; volume 1, p. 428-432. http://dx.doi.org/10.1109/NSSMIC.2005.1596286 , Nov-2005